

Pathway towards the faintest gas emission detection: myth, obstacles, and breakthroughs

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ABSTRACT

Detecting faint gaseous structures in the circumgalactic and intergalactic medium (CGM/IGM) in emission demands sensitivities that exceed those of current and planned missions. A common misconception is that “the bigger, the better,” but for diffuse sources the more relevant principle is “the faster, the better”: systems with lower focal ratios (F-numbers) and larger étendue ($A\Omega$) achieve higher sensitivity. However, practical limits on focal ratio, aberration control, detector noise, and stray light impose significant challenges. In this work, we analyze these limitations and assess the feasibility of a novel multi-aperture telescope–spectrograph architecture in which individual telescopes feed co-pointed fibers at the focal planes into a common spectrograph. Concept-level modeling shows that an array of 100 telescopes with co-pointed fibers can achieve a sensitivity of $\sim 10^{-21}$ erg s⁻¹ cm⁻² arcsec⁻² at 200 nm in a less than 100-hour exposure, nearly 1,000 times fainter than previously observed ultraviolet CGM line emission. The concept system could be realized within a volume limitation of standard SmallSat form-factor, with existing technologies for the payload. This study demonstrates that a multi-aperture system provides a viable and scalable pathway toward the unprecedented sensitivity required to probe the faintest diffuse structures in the CGM/IGM.

Keywords: Circumgalactic medium, Intergalactic medium, Ultraviolet astronomy, Faint emission detection, Multi-aperture telescopes, Etendue, Spectrographs

1. INTRODUCTION

Mapping the circumgalactic medium (CGM) and intergalactic medium (IGM) in emission is a key objective for understanding how galaxies grow and evolve over cosmic time. The CGM and IGM together comprise the majority of a galaxy’s baryonic mass budget, and the flows of material into and out of galaxies regulate star formation and drive galaxy evolution.^{1,2}

Traditionally, studies of the CGM/IGM have relied on absorption-line spectroscopy of bright background objects such as quasars. In this technique, light from the background source is absorbed by intervening gas, producing absorption features in its continuum that reveal the physical and chemical properties of the CGM/IGM. While powerful, this method has a fundamental limitation: it only probes regions along the sightlines to rare bright background objects, and thus cannot map the spatial distribution of diffuse gas.

Observations of the CGM/IGM in emission provide a crucial complement to absorption studies. Unlike absorption, which probes only discrete sightlines to background sources, emission measurements directly trace the spatial distribution of the gas without requiring bright background objects. However, the emission is extraordinarily faint and diffuse, making detection challenging. This difficulty is compounded by the fact that many of the strongest diagnostic transitions—such as Lyman- α , O VI, and C VI—lie in the rest-frame ultraviolet (UV), necessitating space-based instrumentation for direct detection and mapping. Observing these transitions near their rest-frame UV wavelengths is generally more favorable than targeting the redshifted lines in the visible, since the surface brightness of diffuse emission dims rapidly with redshift, scaling as $(1+z)^{-4}$.

Despite the importance of observing those CGM lines in emission, there remains a significant gap between the capabilities of past or current missions and the desired sensitivity required to detect the faintest CGM/IGM emission. The expected surface brightness of diffuse UV lines is of order 10^{-20} erg s⁻¹ cm⁻² arcsec⁻², and in many cases may extend to 10^{-21} erg s⁻¹ cm⁻² arcsec⁻².² This is two to three orders of magnitude fainter than the most sensitive CGM/IGM detections to date. For example, direct spectroscopic detection of far-UV O VI line with the *FUSE* satellite have been made in only four nearby galaxies, at surface brightness levels of $\sim 10^{-18}$ erg

$\text{s}^{-1} \text{cm}^{-2} \text{arcsec}^{-2}$.³⁻⁵ On the ground, redshifted Lyman- α emission from the IGM has been detected at $z \sim 1$,⁶ also at the $\sim 10^{-18} \text{erg s}^{-1} \text{cm}^{-2} \text{arcsec}^{-2}$ level. The upcoming *Aspera* mission aims to reach $\sim 10^{-19} \text{erg s}^{-1} \text{cm}^{-2} \text{arcsec}^{-2}$ sensitivity,⁷⁻⁹ but this remains at least two orders of magnitude brighter than the levels the community ultimately seeks to achieve.

Table 1. Signal counts for point and diffuse sources.

Source Type	Point Source		Diffuse Source	
Telescope Diameter	10 cm	100 cm	10 cm	100 cm
F/#	15			
Resolution element size	2 × 2 pixels			
Flux or Surface Brightness	1 × 10 ⁻¹⁵ erg s ⁻¹ cm ⁻²		1 × 10 ⁻¹⁵ erg s ⁻¹ cm ⁻² arcsec ⁻²	
Exposure time	10 hours			
Signal Counts (e ⁻)	92	9,200	1,000	1,000

1 pixel = $12 \mu\text{m}$

Assuming monochromatic light at 645 nm

Single photon energy at 645 nm = $3.079 \times 10^{-12} \text{erg}$

System throughput: 10%

2 pix = $24 \mu\text{m}$ = 3.3 arcsec (F/15, 10 cm diameter) or 0.33 arcsec (F/15, 100 cm diameter)

In this proceeding, we briefly review the myths, obstacles, and possible solutions relevant to the detection of the faintest CGM/IGM emission. First, we address the common misconception in telescope and instrument design that “bigger is better,” clarifying the metrics most relevant for diffuse-source sensitivity. Second, we summarize the primary technical obstacles to detecting faint and diffuse emission, including limitations from optics, detectors, and stray light. Lastly, we present a concept-level instrument architecture—based on a multi-aperture approach—that leverages current technologies to realistically approach the required sensitivity for detecting the faintest CGM/IGM emission.

2. BREAKING THE MYTH: BIGGER TELESCOPES ARE NOT NECESSARILY ADVANTAGEOUS FOR FAINT, DIFFUSE EMISSION SOURCES

A common misunderstanding in telescope design for faint sources is the idea that “the bigger, the better.” In other words, it is often assumed that a larger primary aperture will always provide greater sensitivity. While this is true for point sources, it is not necessarily the case for diffuse emission.

For point sources, the collected signal increases with the square of the aperture diameter, D^2 . Parallel rays from a distant source are focused into a point spread function (PSF) at the focal plane. Assuming the PSF remains smaller than a resolution element (e.g., 2×2 pixels), the number of detected photons scales directly as D^2 . The counts are given by

$$N_{e^-} \propto F_{\text{point}} \times A_{\text{eff}} \times t \propto F_{\text{point}} \times D^2 \times \epsilon \times t, \quad (1)$$

where F_{point} is the source flux [$\text{erg s}^{-1} \text{cm}^{-2}$], A_{eff} is the effective collecting area, ϵ is the system throughput, and t is the integration time.

For diffuse sources, however, the situation is very different. Although the aperture increases the collecting area as D^2 , the angular size corresponding to a fixed detector element decreases as D^{-2} . The net result is that the telescope diameter cancels out, leaving the signal dependent only on the optical speed ($F/\#$):

$$N_{e^-} \propto S_{\text{diffuse}} \times A_{\text{eff}} \times \Omega_{\text{pix}} \times t \propto S_{\text{diffuse}} \times (F/\#)^{-2} \times \epsilon \times w^2 \times t, \quad (2)$$

$$\Omega_{\text{pix}} = \left(\frac{1}{f}\right)^2 \times w^2 = \left(\frac{1}{F/\# \times D}\right)^2 \times w^2, \quad (3)$$

where S_{diffuse} is the surface brightness [$\text{erg s}^{-1} \text{ cm}^{-2} \text{ arcsec}^{-2}$], f is the focal length, w is the resolution element size, and Ω_{pix} is its solid angle. Thus, for diffuse emission, the relevant metric is the focal ratio, not the primary aperture: “*The faster the better,*” not “*The bigger the better.*” A simple numerical demonstration is provided in Table 1, and readers may explore this effect further using the online calculator at signalcounts.astrohchung.com.

It is important to note, however, that larger apertures still offer two important advantages. First, they provide improved angular resolution, which is crucial for resolving fine spatial details. Second, for emission that is not fully homogeneous—such as compact knots or filamentary structures—a larger aperture can reveal features that would remain undetectable to an instrument with the same sensitivity but lower spatial resolution. Nevertheless, when observing truly uniform diffuse emission, increasing aperture size alone does not increase photon counts at fixed $F/\#$. In such cases, optimizing the optical speed and system throughput remains the more effective strategy for pushing sensitivity limits.

3. OBSTACLES TO FAINT EMISSION DETECTION: COST, COMPLEXITY, THROUGHPUT, AND NOISE

Achieving sensitivity to faint, diffuse emission in the CGM/IGM requires overcoming multiple technological challenges. Table 2 summarizes key enabling parameters for a telescope–spectrograph system optimized for diffuse sources. Four categories stand out as the most difficult to advance simultaneously: (1) fast optics, (2) high throughput, (3) low detector background, and (4) stray light control. Here we focus on the ultraviolet regime, where the development and diversity of component-level technologies lag significantly behind those available at optical and infrared wavelengths. Although our discussion emphasizes telescope–spectrograph systems, many of these considerations also apply to imaging instruments. Community-wide efforts, particularly through NASA technology development programs, are actively addressing all of these fronts.

3.1 Fast optics

As shown in Section 2, fast optical systems maximize photon collection from diffuse sources at fixed aperture diameter. Designing fast optics, however, is intrinsically difficult. Faster systems with wide field-of-view often require multiple surfaces and aspheric or freeform elements to control aberrations. The point-spread function (PSF) degrades rapidly with defocus, and fabrication and alignment tolerances become increasingly tight. Each added surface also reduces throughput. Together, these factors drive higher cost and complexity.

Maintaining a small PSF is especially important for diffuse emission, since background noise (detector, sky, or stray light) scales with the footprint of the resolution element on the detector. Thus, optical quality and small PSF size are essential not only for image sharpness but also for minimizing noise.

3.2 High throughput

System throughput depends on the reflectivity of coatings, the efficiency of gratings, and the quantum efficiency of detectors. In the UV, each presents unique challenges.

Coatings: Between 90–120 nm, coating options are limited. Standard aluminum-based coating ($\text{Al}+\text{MgF}_2$), as used on *HST*, has poor reflectivity in this range. $\text{Al}+\text{LiF}$ provides $>50\%$ reflectivity near 103 nm but degrades rapidly in humidity due to the hygroscopic nature of LiF.^{10,11} Nevertheless, advanced Far-UV coatings such as enhanced LiF capped with MgF_2 ($\text{eLiF}+\text{MgF}_2$) or XeLiF coatings are being developed, providing less hygroscopic characteristics while achieving $>70\%$ reflectivity at 103 nm^{10–14}

Gratings: Achieving high diffraction efficiency requires precise groove profiles optimized for a given order and wavelength. In the UV, required groove densities can exceed 10,000 lines/mm,¹⁵ pushing fabrication to technological limits. Additional challenges arise from the need for non-parallel grooves, curved substrates, and multi-layer coatings, all of which complicate both manufacture and performance modeling. Recent progress in silicon-based gratings has shown promise for combining high efficiency with low scatter, while enabling dense groove patterns, curved geometries, and non-parallel features.^{16–22} Continued development of this technology is expected to play a critical role in advancing UV spectrographs for faint-source detection.

Detectors: UV quantum efficiency is improving steadily. For silicon CCD/CMOS devices, JPL’s delta-doping technique has produced high-efficiency detectors,²³ often combined with band-specific coatings.^{24,25} For

Table 2. Key challenging parameter spaces for the instrument design of the faint diffuse emission detection.

Fast Optics	Aberration control difficulty More # of surfaces High complexity (Asphere, freeform) High cost
High Throughput	High coating reflectivity High diffraction efficiency High detector quantum efficiency
Low Detector Noise	Low intrinsic background (MCP) Low dark current (CCD/CMOS) Low readout noise (CCD/CMOS)
Low Straylight	Low surface roughness mirror Low surface roughness grating UV sky background knowledge UV airglow line knowledge

microchannel plates (MCPs), atomic-layer-deposited photocathodes significantly improve QE over previous generations.^{26,27}

3.3 Low detector noise

Detecting faint diffuse emission requires Poisson-limited performance, which in turn demands suppressing detector-induced noise. Microchannel plates (MCPs) offer an intrinsic advantage: they are true photon-counting devices with negligible readout noise and very low intrinsic background (<0.1 counts $\text{s}^{-1} \text{cm}^{-2}$, corresponding to $\sim 10^{-7}$ $\text{e}^- \text{pix}^{-1} \text{s}^{-1}$ for $10 \mu\text{m}$ pixels). However, the effective background rate of MCPs has been shown to increase approximately logarithmically with the total spacecraft mass,²⁸ an effect that must be considered in mission design and noise budgeting.

Modern silicon detectors (EMCCD, Skipper CCD) can also operate in near-photon-counting regimes. Readout noise can be reduced by on-chip binning in regular CCDs, though CMOS detectors suffer from pixel-dependent noise that scales with array size. Dark current in silicon devices requires cooling, though recent CMOS arrays achieve very low dark current even above 273 K.

3.4 Low stray light

Stray-light suppression is critical for faint-source detection, since uncontrolled scattered light can overlap directly with the science signal on the detector and contribute significantly to the total noise budget. Effective stray-light control must therefore be treated as a primary design driver, rather than a secondary consideration, in instruments aiming to detect diffuse emission at UV wavelengths.

Optical surfaces: Scattered light from optical surfaces scales with the square of the surface roughness and inversely with the square of the wavelength, as described by the Harvey–Shack scatter model for Total Integrated Scatter (TIS).²⁹ This scaling suggests sub-0.1 nm roughness for UV mirrors. While such mirrors can be fabricated, the associated cost is very high. Diffraction gratings present an even greater challenge: in-band scatter propagates along the dispersion direction and cannot be baffled, making low-scatter fabrication essential. In addition, light from non-operating diffraction orders such as the 0th order may introduce significant background on the detector. This issue is especially important in fast optical systems, where designing effective baffles is non-trivial and order suppression must be carefully engineered into the grating and system design.

Astrophysical background: The cosmic ultraviolet background, though faint, has an estimated upper bound of $\sim 10^{-19}$ $\text{erg s}^{-1} \text{cm}^{-2} \text{\AA}^{-1}$, which can become significant relative to the expected surface brightness of

diffuse emission.^{30,31} Its impact on the total noise budget depends strongly on the instrument configuration and spectral resolution. Because this background continuum is always overlapping with the science signal, it must be accurately characterized and included in sensitivity estimates. Failure to account for it can lead to substantial errors in exposure time predictions and observation planning.

Airglow: In low Earth orbit, the UV sky is dominated by bright atmospheric airglow lines.^{32–36} Although these are intrinsically narrow spectral features, scatter from grating surfaces redistributes their light and effectively acts as a continuum background on the detector. The extreme brightness of UV airglow therefore places strict requirements on out-of-field light suppression, since the entire sky may contribute. In addition, airglow strength is highly variable, depending on the relative geometry of the Sun, Earth, and spacecraft, as well as the instrument boresight angle relative to the nadir. Accurate knowledge of these time-, geometry-, and attitude-dependent variations is essential for planning observations of faint diffuse emission from low Earth orbit.

4. AN ARCHITECTURAL SOLUTION: SUPERGRASP ARRAY IN LEO

We propose an architectural solution, the *SuperGrasp Array*, designed to enable detection of the faintest diffuse emission from the CGM/IGM. The concept consists of one hundred (or more) identical small telescopes, each coupled to optical fibers that feed one or more spectrographs. Although one hundred telescopes may sound excessive, the compact aperture size (5 cm diameter) makes such an array feasible within the mass and volume constraints of a SmallSat platform, while remaining within a reasonable budget.

4.1 Design motivation

The SuperGrasp Array is designed to reach a surface-brightness sensitivity of $\sim 10^{-21}$ erg s⁻¹ cm⁻² arcsec⁻² in a less than 100-hr exposure. This corresponds to a factor of $\sim 1,000$ improvement over the faintest UV detections of CGM emission to date. The concept builds directly on the demonstrated progress of the *Aspera* SmallSat mission, which will soon achieve sensitivities of $\sim 10^{-19}$ erg s⁻¹ cm⁻² arcsec⁻² in the far-UV (O VI line at around 103–104 nm).

Aspera, a NASA Astrophysics Pioneer mission, represents a breakthrough in UV instrumentation through: (1) a fast optical system (F/2.2), (2) minimal reflections (one mirror and one grating), (3) a photon-counting MCP detector, (4) comprehensive stray-light suppression, and (5) multiple (two) parallel channels.^{7–9} These enabling design choices demonstrate that unprecedented faint-source sensitivity is achievable when system parameters are optimized. Building on this foundation, we recognize that scaling the number of independent optical systems provides a practical and near-term path to reach the additional orders of magnitude in sensitivity required for probing the faintest diffuse CGM/IGM emission.

4.2 System design concept

The central idea of the SuperGrasp Array is to increase the number of telescope apertures in order to maximize the photon-collecting capability of the system. This approach is inherently scalable and is designed to minimize per-aperture manufacturing costs. Alignment strategy considerations have also been incorporated at this conceptual stage. The system is optimized for space application in the UV, where the astrophysical background is very low.

The idea of using multiple apertures to increase étendue is not entirely new; existing or developing multi-aperture ground/space facilities such as Dragonfly, LFAST, and Nautilus.^{37–39} However, the SuperGrasp concept is unique in being tailored specifically to detect faint, diffuse UV line emission, where the advantages of the multi-aperture approach are maximized by combining a low-background environment with Poisson-limited sensitivity. The architecture consists of three main elements: (1) an array of off-axis parabolic mirrors as telescopes, (2) a small number of fibers at each telescope focal plane, and (3) one or more fiber-fed spectrographs that produce spectra from the fiber outputs.

The number of telescopes is the key parameter enabling large photon-collecting power. As discussed in Section 2, for diffuse sources the photon-collection rate depends on the focal ratio ($F/\#$), not on the individual aperture size. Consequently, the total collecting power of the array scales linearly with the number of telescopes. However, the aperture diameter cannot be reduced arbitrarily. With a fixed sampling element (e.g., a fiber core), the angular resolution is set by the focal length, $f = F/\# \times D$. If D is too small, the angular resolution will no

longer satisfy the science requirements. For this reason, each telescope is designed around an off-axis parabolic mirror: this configuration minimizes the number of reflections, eliminates pupil obstruction, and provides high PSF quality near the optical axis, even though the usable field of view is limited. All telescopes and their associated fibers are co-aligned, ensuring that every fiber simultaneously samples the same field of view on the sky.

Light collected by each telescope is coupled into one or more optical fibers at the focal plane. Modern UV fibers provide reasonable transmission at wavelengths longer than ~ 190 nm. The fiber interface allows the multi-aperture array to be fully exploited without requiring each telescope to be directly connected to its own spectrograph. Instead, light from all telescopes can be routed to one or more centralized fiber-fed spectrographs. This approach has a strong heritage in large ground-based facilities, including the Sloan Digital Sky Survey spectrographs,⁴⁰ Hectospec on the MMT,⁴¹ and AAOmega on the Anglo-Australian Telescope.⁴²

The fiber-fed spectrographs collimate, disperse, and refocus the fiber outputs onto detector arrays. Although fiber-fed designs are widely used, their development still requires careful optimization, particularly for fast optical systems. A fundamental trade-off exists between wavelength coverage and spectral resolution: attempting to maximize both reduces throughput, increases optical complexity, and drives larger instrument and detector formats, which in turn raise cost.

One possible strategy to reduce spectrograph cost is to minimize the spacing between fibers at the spectrograph entrance. While this normally increases spectral cross-talk, in the present application the effect is less problematic if adjacent fibers sample the same field of view on the sky. For example, if each telescope feeds seven fibers, then a slit could be arranged into seven groups, each containing 100 adjacent fibers (one from each telescope). In this configuration, cross-talk between fibers within a group would not affect the science, since all fibers sample the same sky region.

Depending on instrument design, additional multiplexing could be achieved by placing multiple fiber slits in the spectral direction, combined with bandpass filters on the spectrograph to mitigate spectral overlap. In this case, emission lines from different bandpasses would remain separable even with multiple slits. Such approaches could significantly increase multiplexing efficiency and enable many more fibers to be accommodated per spectrograph unit by utilizing the full detector area as much as possible.

An important point is that, for diffuse emission, the total number of collected photons depends on the number of resolution elements rather than the physical size of the primary mirror (Section 2). For example, a single 100-cm F/2 telescope with 100 fibers at its focal plane collects the same number of photons as an array of 100 F/2 10-cm telescopes with one fiber each, assuming identical throughput. In this sense, the light-collecting power is effectively defined by the number of fibers to be accommodated by the spectrograph.

Nevertheless, the multi-aperture approach offers clear advantages in mass and volume, both of which are critical for space application. For example, a 100-cm monolithic telescope would require fiber runs that are several meters longer than those in a distributed 10-cm array, introducing handling challenges and additional UV throughput loss. In addition, when photons from multiple adjacent fibers in a large single-aperture system are integrated, the effective spatial resolution degrades to that of the smaller telescope, losing the advantage of higher spatial resolution. A realistic trade study must therefore weigh alignment complexity against the savings in mass and volume, as well as the resulting science performance.

4.3 Example system design

Here we describe an example system design for a multi-aperture telescope designed to detect faint diffuse emission lines using existing technologies. The array consists of 100 circular, F/2.2 off-axis parabolic telescopes, each 5 cm in diameter. At the focal plane of each telescope, seven high-NA fibers matched to F/2.2 are arranged in a hexagonal pattern. Each fiber has a 50 μm core, corresponding to a sky coverage of 93.8 arcsec in diameter. In total, the 700 fibers feed light into one or more fiber-fed spectrographs.

Each spectrograph employs conventional collimator-camera optics, a diffractive grating, and a photon-counting EMCCD detector. The chosen wavelength range of 190–220 nm targets redshifted CGM emission lines, while also enabling the use of proven Al+MgF₂ coatings and UV-transmissive fibers. The telescope assembly (excluding spectrographs) occupies less than $60 \times 60 \times 20$ cm³, well within the constraints of a SmallSat-class spacecraft bus.

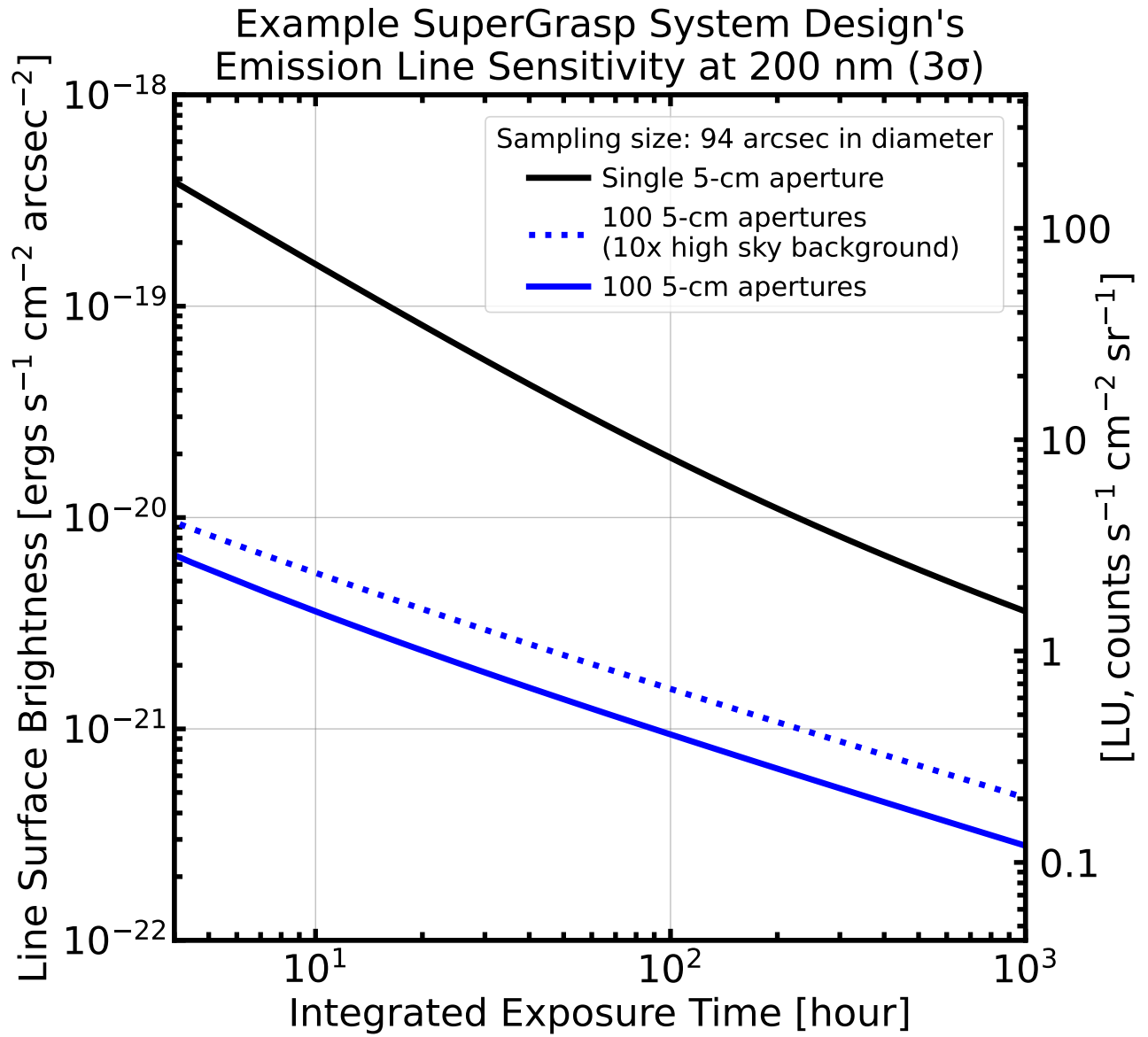


Figure 1. Projected line-emission sensitivity of the SuperGrasp array concept. The black curve shows the sensitivity of a single 5-cm aperture, while the blue curves show the gain from 100 co-aligned apertures. The array achieves a surface brightness sensitivity of $\sim 10^{-21} \text{ erg s}^{-1} \text{cm}^{-2} \text{arcsec}^{-2}$ in less than 100-hour ($S/N=3$).

4.4 Projected performance

The projected sensitivity of the example SuperGrasp array is shown in Figure 1. Assumptions for the performance estimate include:

- One telescope reflection; five reflections in the spectrograph (including the grating).
- 85% per reflection, based on Al+MgF₂ coating performance with margin.
- Grating efficiency: 65%.
- Detector quantum efficiency: 65%.
- Fiber transmission (including input/output losses): 75%.

These values yield a total system throughput of $\sim 12\%$. The effective area is 2.35 cm^2 per fiber per telescope, corresponding to 235 cm^2 for 100 co-aligned telescopes.

The baseline detector is an EMCCD (1 pixel = $13 \text{ }\mu\text{m}$) with a dark current of $0.01 \text{ e}^- \text{ pix}^{-1} \text{ hr}^{-1}$ and an effective read noise of $0.01 \text{ e}^- \text{ pix}^{-1}$ per read (RMS). A UV continuum background of $10^{-21} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ arcsec}^{-2} \text{ \AA}^{-1}$ is adopted, with a “high-background” case of $10^{-20} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ arcsec}^{-2} \text{ \AA}^{-1}$. The fiber PSF is assumed to cover 4×4 pixels, based on a $50 \text{ }\mu\text{m}$ fiber core and a spectrograph with no magnification. This footprint corresponds to 0.1 nm in dispersion direction, or $R \sim 2000$ at 200 nm . The readout cadence (single exposure time) is set to 900 s , consistent with continuous target visibility in low-Earth orbit.

Under these assumptions, the SuperGrasp system achieves a surface brightness sensitivity of $\sim 10^{-21} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ arcsec}^{-2}$ in a 89.55 hours of exposure, enabling a factor of $1,000$ improvement over the faintest UV CGM emission observed to date.

5. SUMMARY

We have presented a framework for developing instruments capable of detecting extremely faint, diffuse ultraviolet CGM/IGM emission lines, along with a concept-level solution that could achieve this goal.

First, we addressed the common misconception that larger apertures alone improve surface brightness sensitivity. For diffuse sources, sensitivity depends instead on the optical speed ($F/\#$) of the system. We then identified the key design parameters that must be optimized to reach the required sensitivities: fast optics, high throughput, low detector noise, and rigorous stray-light control.

Finally, we introduced the *SuperGrasp* concept: a multi-aperture, high-grasp space telescope array coupled to fiber-fed spectrographs. Using currently available technologies, this architecture can achieve surface brightness sensitivities of $\sim 10^{-21} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ arcsec}^{-2}$ at $\sim 200 \text{ nm}$ in less than 100-hour exposure—roughly $1,000$ times fainter than previous UV detections of CGM/IGM emission and aligned with the science goals identified in the Astro2020 Decadal Survey. An example implementation, consisting of 100 off-axis parabolic telescopes each 5 cm in diameter, fits within the mass and volume constraints of a SmallSat-class platform.

The SuperGrasp approach is scalable and versatile. With appropriate mission design and science definition, the same architecture could be adapted to a range of astrophysical investigations beyond CGM/IGM studies, providing a practical pathway toward unprecedented sensitivity in faint-source UV astronomy.

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